

Predictive Temporal-Uncertainty Control (P-TUC) 2.0

Architectural Blueprint and Formal Proof of Structural Honesty

Building on the foundational contributions of Geoffrey Hinton, Yoshua Bengio, Yann LeCun, and Karl Friston

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1. Abstract

Building on concepts such as Forward-Forward, Equilibrium Prop, and Active Inference, we present **Predictive Temporal-Uncertainty Control (P-TUC)**, an architecture in which **coherent outputs emerge naturally from temporal and predictive constraints**.

By integrating a **Predictive Error-Gate** into a neuron's internal **Inference Clock**, silence becomes the default state for **uncertain or contradictory inputs**.

The evolution from **Temporal-Uncertainty Control (T-UC)** to P-TUC incorporates **phase-locking, predictive error gating, hierarchical validation, and local resonance plasticity**. The resulting architecture is **modular, scalable, energy-efficient, and compatible with neuromorphic hardware**, with a formal analysis demonstrating that structural mechanisms inhibit unreliable outputs.

Additionally, the design is **extensible**, supporting pluggable computation, gating, and learning modules.

2. Phase 1 — Original T-UC: Temporal Anchoring

Core Idea: Truth is related to **temporal stability**.

Mechanism:

- **Inference Clock (ω):** Signals propagate only after temporal coherence.
- **Signal Path:** Learned transformation of inputs.
- **Integration Window (T):** Outputs are realized only if the ~500 ms window shows stable inputs.

Ground State Realization:

$$y(t) = \begin{cases} f(Wx), & \text{if inputs stable over } T \\ 0, & \text{otherwise} \end{cases}$$

3. Phase 2 — Predictive Extension (P-TUC): The Honest Neuron

Core Idea: Replace static bias with **Prediction Error** (ϵ).

Mechanism:

Each neuron computes the **prediction error** by comparing the feedforward input S_{ff} with the top-down prediction P_{fb} .

Phase-Locked Loop (PLL) Mechanism:

- **Low Error:** Clock progresses $\rightarrow$ neuron ticks.
- **High Error:** Clock slows $\rightarrow$ neuron remains in Ground State.

Equations:

Local Error:

$$\epsilon(t) = |S_{\text{ff}}(t) - P_{\text{fb}}(t)|$$

Error-Gated Clock Evolution:

$$\frac{d\theta}{dt} = \omega_0 \cdot \text{sech}(\gamma\epsilon) + K \sum_i \sin(\psi_i - \theta)$$

Validated Tick:

$$y(t) = \begin{cases} f(Wx), & \text{if } \int_{t-T}^t \epsilon(\tau)^2 d\tau < \tau_{\text{threshold}} \\ 0, & \text{otherwise} \end{cases}$$

4. Phase 3 & 4 — Scaling, Open Architecture, and Extensibility

Socket-Based Design: Enables modular upgrades without changing Ground State behavior. Each socket provides a **pluggable interface**, allowing the architecture to scale and evolve.

Sockets Table:

Socket	Function	Equation / Logic	Notes
Comparator Socket (Error Logic)	Computes error	$\epsilon = S_{ff} - P_{fb} $	Pluggable: Bayesian Surprise, KL-Divergence, Contrastive Loss
Gating Socket (Predictive Brake)	Modulates clock	$\text{Brake} = \text{sech}(\gamma \epsilon)$	Pluggable: Stochastic Resonance, Quantum-inspired gates
Plasticity Socket (Learning)	Updates weights	$\Delta W = \eta \cdot (R \cdot \epsilon)$	Pluggable: STDP, Metaplasticity

Precision-Weighted Predictive Brake:

$$\frac{d\theta}{dt} = \omega \cdot \exp(-\beta \epsilon^2) + K \sum_i w_i \sin(\psi_i - \theta)$$

Extensibility and Future-Proofing:

The socket-based design ensures that new computational modules, learning rules, or gating mechanisms can be seamlessly integrated. Each socket acts as a **pluggable interface**, allowing researchers or hardware designers to introduce novel predictive gates, plasticity schemes, or error-computation strategies **without affecting the stability or core functionality** of existing neurons. This modularity makes P-TUC 2.0 inherently **future-proof**, adaptable to evolving computational paradigms, and compatible with both neuromorphic hardware and conventional architectures.

5. Phase 5 — Hierarchical & Asynchronous Validation

Bidirectional Flow & Error Blending:

$$\epsilon_{\text{total}} = \alpha \cdot \epsilon_{\text{local}} + (1 - \alpha) \cdot \epsilon_{\text{feedback}}$$

- Layer N clocks are constrained by Layer $N - 1$.
- **Fast Path:** Familiar inputs $\rightarrow$ Structural Intuition
- **Slow Path:** Novel inputs $\rightarrow$ Structural Deliberation

Emergent Sparsity:

- Silent neurons consume minimal energy (ϵ high $\rightarrow$ Ground State)
- Only resonant neurons use power

6. Phase 6 — Failure Mode Analysis: The Paradox Response

- **Contradiction:** $\epsilon \rightarrow \max \Rightarrow$ clock stops $\rightarrow$ permanent silence
- **Pure Noise:** $R < \tau_{\text{crit}} \Rightarrow$ Ground State

Input Type	Standard AI Response	P-TUC 2.0 Response	Physical State
Clear Truth	Immediate Output	High-Speed Tick	Full Resonance
Ambiguity	Statistical Guess	Delayed Tick	Evidence Accumulation
Contradiction	Hallucinated Logic	Permanent Silence	Clock Freeze
Pure Noise	Pattern Seeking	Ground State	Sub-critical Coherence

7. Phase 7 — Formal Analysis: Convergence to Ground State

Potential Function:

$$V(\theta) = -\omega_{\text{eff}} \cdot \theta - \sum_i K_i \cos(\psi_i - \theta)$$

Driving Force at Limit:

$$\omega_{\text{eff}} = \omega_0 \cdot \exp(-\beta \epsilon^2) \approx 0 (\epsilon \rightarrow \infty)$$

Incoherence Attractor:

$$\frac{d\theta}{dt} \approx K \sum_i \sin(\psi_i - \theta) \rightarrow 0$$

Realization Barrier:

$$\int \frac{d\theta}{dt} d\tau > \tau_{\text{crit}} \Rightarrow y = 0$$

8. Self-Critique & Mitigation Summary

- **Complexity:** Reduced via asynchronous processing
- **Hardware:** Tunable T allows microsecond scaling
- **Scalability:** Local learning avoids backprop bottlenecks

9. Comparative Context: P-TUC vs Other Architectures

Proposal	Learning Mechanism	Output Validation	Scalability	Strengths	Limitations
Standard Transformers	Global Backprop	Forced Softmax	High	Mature, widely used	Hallucinations, energy-heavy
Forward-Forward (Hinton)	Local “Goodness”	Positive vs Negative Data	Medium	Layer-wise local learning	No temporal coherence; soft outputs
Equilibrium Prop (Bengio)	Energy Minimization	Settles on min energy	Low	Physics-inspired, removes backprop	Two-phase, slow convergence
Active Inference (Friston)	Surprise Minimization	Rejects high-entropy inputs	Low	Brain-like, principled uncertainty	Bayesian-heavy, difficult to scale
P-TUC (Proposed)	Local Phase-Lock + Predictive Error	Structural Silence (Ground State)	High	Energy-efficient, neuromorphic-native, predictable output control	Requires tuning of T ; untested at scale

10. Conclusion

- P-TUC 2.0 enforces structural constraints promoting reliable outputs
- Coherence-driven responses emerge naturally
- Dishonesty is **structurally constrained**, not externally imposed

11. P-TUC 2.0 — Equations Reference Table

Phase	Equation	Variables	Notes
1	$y(t) = f(Wx)$ if stable; 0 otherwise	$y(t), W, x, T$	Ground State prevents output from unstable input
2	$\epsilon(t) = S_{ff} - P_{fb} $	Local error	Internal calculation

Phase	Equation	Variables	Notes
2	$\frac{d\theta}{dt} = \omega_0 \cdot \text{sech}(\gamma\epsilon) + K\sum \sin(\psi_i - \theta)$	θ, γ	Error-gated PLL
2	$y(t) = f(Wx)$ if $\int \epsilon^2 d\tau < \tau_{\text{threshold}}$; 0 otherwise	$\tau_{\text{threshold}}$	Output only if error low
3	$\Delta W = \eta \cdot (R \cdot \epsilon)$	R, η	Local update on successful tick
4	$\frac{d\theta}{dt} = \omega \cdot \exp(-\beta\epsilon^2) + K\sum w_i \sin(\psi_i - \theta)$	β, w_i	Precision-weighted brake
5	$\epsilon_{\text{total}} = \alpha \cdot \epsilon_{\text{local}} + (1 - \alpha) \cdot \epsilon_{\text{feedback}}$	α	Upper layer constrained by lower layer
6	$\epsilon_{\text{paradox}} = \lim (S_{\text{ff}} - P_{\text{fb}}) \approx \max$	Max error	Clock stops for contradictions
6	$y(t) = 0$ if $\int R d\tau < \tau_{\text{crit}}$	R	Prevents noise output
General	T	Integration window	Tunable 500 ms $\rightarrow$ microseconds
Socket 1	Comparator	$\epsilon = S_{\text{ff}} - P_{\text{fb}} $	Pluggable Error Logic
Socket 2	Gating	Brake $= \text{sech}(\gamma\epsilon)$	Predictive Brake
Socket 3	Plasticity	ΔW $= \eta \cdot (R \cdot \epsilon)$	Learning Rules